

Jakub (Kuba) Preis

kuba.c.preis@gmail.com

Academic & Research Appointments

Assistant Professor

Department of Mechanical Engineering, Cal Poly SLO

August 2026 - Present

San Luis Obispo, CA

- Responsible for teaching design courses and conducting research in metal additive manufacturing.

Industry R&D Experience

Research and Development Engineer

Valcun

October 2024 - July 2026

Ghent, Belgium

- Part of a team of 5 engineers developing a novel aluminum additive manufacturing technology.
- Responsible engineer for design, development, and testing of hardware and PLC programming (i.e. system control and modeling).

Additive Manufacturing Engineer

Divergent Technologies Inc.

June 2020 - December 2020

Lübeck, Germany

- Responsible for build preparation, machine operation, and testing of parts printed using a prototype 12 laser laser powder bed fusion machine.
- Submitted recommendations to laser powder bed fusion machine designers with the goal of improving end user experience.

Additive Manufacturing Engineer

Divergent Technologies Inc.

August 2019 - June 2020

Los Angeles, CA

- Developed a statistical laser powder bed fusion machine health model using C++ database software.
- Prepared parts for additive manufacturing using Materialise Magics (build orientation and support addition).
- Conducted R&D experiments to quantify additive manufacturing design constraints and document results.

Test and Validation Engineering Intern

Divergent Technologies Inc.

June 2018 - September 2018

Los Angeles, California

- Worked on testing 3D printed aluminum suspension components.

Manufacturing Process Improvement Engineering Intern

Plastikon Industries Inc.

June 2017 - September 2017

Hayward, CA

- Worked on the recently released Tesla Model 3 HVAC unit and center console manufacturing lines.

Education

Doctor of Philosophy in Mechanical Engineering

Oregon State University

January 2021 - August 2024

- Main Project: Developing metallurgical pathways for the defect free fabrication of Inconel - Cu alloy multi-metal components via additive manufacturing.
- Advisor: Dr. Somayeh Pasebani

Bachelor of Science in Manufacturing Engineering

California Polytechnic State University, San Luis Obispo

September 2015 - June 2019

- Main Project: Redesign, validation, and printing of a 1U cubesat frame using laser powder bed fusion.
- Advisor: Dr. Xuan Wang

Publications

Preis, J., Lawson, S.B., Wannenmacher, N., Pasebani, S., 2025, Joining Inconel 718 and GRCo42: A framework for developing transition compositions to avoid cracking and brittle phase formation. *Materials & Design*, 252, p.113733.

Preis, J., Lawson, S.B., Lee, I., Kawasaki, M., Bay, B.K., Manoharan, S., Paul, B.K. and Pasebani, S., 2024. Influence of travel speed on microstructure and mechanical behavior of Inconel 625 fabricated using wire fed laser directed energy deposition. *Journal of Materials Processing Technology*, p.118464.

Preis, J., Wang, Z., Howard, J., Lu, Y., Wannenmacher, N., Shen, S., Paul, B.K. and Pasebani, S., 2024. Effect of laser power and deposition sequence on microstructure of GRCo42-Inconel 625 joints fabricated using laser directed energy deposition. *Materials & Design*, 241, p.112944.

Preis, J., Xu, D., Paul, B.K., Eschbach, P.A. and Pasebani, S., 2024. Effect of Liquid Miscibility Gap on Defects in Inconel 625–GRCo42 Joints through Analysis of Gradient Composition Microstructure. *Journal of Manufacturing and Materials Processing*, 8(1), p.42.

Zardoshtian, A., Rezaei, A., Esmailizadeh, R., **Preis, J.**, Minasyan, T., Keshavarz, M., Pasebani, S., Jahed, H., Toyserkani, E. Insights into Microstructural Evolution in Functionally Graded Additive Manufacturing of IN625–CuCrZr Alloys: A CALPHAD-Based Thermodynamic Analysis and Experimental Study on the Role of Metastable Miscibility Gap. *Journal of Materials Research and Technology*, 39, p.1612.

Zardoshtian, A., Keshavarz M., Rezaei, A., **Preis, J.**, Minasyan, T., Pasebani, S., Jahed, H., Toyserkani, E. Liquid-state miscibility gap enabled crack suppression in functionally graded additively manufactured Ni-Cu alloy systems. *Journal of Alloys and Compounds*, 1050, p.185705.

Teaching Experience

Teaching Assistant and Laboratory Instructor

January 2023 - March 2023

Oregon State University

Corvallis, OR

- Led laboratory sections for MFGE 337 (Materials and Manufacturing Processes) to provide a hands-on counterpart to the lecture.
- Responsible for developing course material, rubrics, and grading student lab reports.

Guest Lecturer

January 2021 - June 2024

Oregon State University

Corvallis, OR

- Developed content and conducted lectures for the following classes:
 - MFGE 337 (Materials and Manufacturing Processes)
 - MFGE 551 (Additive Manufacturing)

Workshop Instructor

September 2016 - June 2019

Academic Skills Center, California Polytechnic State University, SLO

San Luis Obispo, California

- Taught a 1 unit, bi-weekly workshop in mathematics to supplement lecture course, class size ranging from 10 - 30 students.
- Instructed the following workshops:
 - MATH 128 (Precalculus Algebra Workshop)
 - MATH 151 (Calculus 1 Workshop)
 - MATH 192 (Calculus for Architecture and Construction Management Workshop)

Research Mentoring

Jurre Lefever	February 2026 - June 2026
• Project: Design and fabrication of automated nozzle cleaning solution for aluminum 3D printing.	
Wolf Debaene	February 2026 - June 2026
• Project: Design and fabrication of automated nozzle resurfacing solution for aluminum 3D printing.	
Fran Van Limberghen	February 2025 - June 2025
• Project: Integration of polymer extrusion into existing molten-aluminum additive manufacturing machine.	
Tyler Finch	September 2023 - June 2024
• Project: Characterization of Inconel 718 - GRCop42 mixtures	
Mithrandir Eichner	September 2023 - June 2024
• Project: Characterization of 4043 Al fabricated via Molten Metal Deposition	
Vaishnavee Selvarajoo	September 2020 - June 2022
• Project: Characterizing Cu-Inconel bimetallic joints via Laser Powder Bed Fusion	
Adam Moar	September 2020 - June 2021
• Project: Fabricating Cu-Inconel bimetallic joints via Laser Powder Bed Fusion	

Book

Preis, J., Core Engineering Concepts and Derivations, Manuscript in Preparation.

Description: This book takes a pyramid approach by presenting axioms in the mathematics, applied statistics, physics, and chemistry chapters, and using these to derive fundamental equations within classical mechanics, materials engineering, solid mechanics, thermodynamics, fluid mechanics, heat transfer, and electrical engineering. Work in progress.

Service

Academic journal reviewer for <i>Nature Advanced Manufacturing</i> and <i>Additive Manufacturing</i>	September 2023 - Present
Academic tutor working with foster youth and unhoused populations, San Luis Obispo, CA	September 2016 - June 2019

Conferences

AMPM 2019, Phoenix, Arizona (Presenter)
Westec 2019, Long Beach, California (Attendee)
Formnext 2019, Frankfurt, Germany (Attendee)
TMS 2023, San Diego, California (Presenter)
ICAM 2023, Washington DC, California (Presenter)
Formnext 2023, Frankfurt, Germany (Presenter)
Formnext 2024, Frankfurt, Germany (Attendee)
SFF 2025, Austin, Texas (Presenter)

Other

- Languages:
 - English (Native)
 - Polish (Native)
 - Spanish (professional working proficiency)
 - German (professional working proficiency)
 - Dutch / Flemish (intermediate)
- Hobbies: backpacking, running, biking, traveling